

Scalability & Future Plan

Date	02 July 2026
Team ID	-
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a single machine learning model (Random Forest)	Prediction accuracy may vary for unseen data	High
2	Supports only manual user input	Time-consuming for large numbers of applications	Medium
3	No user authentication or database storage	User records are not stored for future reference	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on cloud with load balancing
2	Data Storage	No database integration	Integrate MySQL or PostgreSQL database
3	Performance	Processes one prediction at a time	Optimize backend and use model caching

4	Security	Basic input validation	Add authentication, HTTPS, and secure API handling
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate a database to store applicant records	1 Month	Improves data management and record keeping
Phase 2	Deploy the application on a cloud platform	2 Months	Enables public access and supports more users
Phase 3	Enhance the prediction model and add user authentication	3 Months	Improves prediction accuracy and application security